Post-operative Cystoid Macular Oedema & Prescribed Medication: 8-Month Retrospective Review

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Introduction

Newmedica Shrewsbury conducted an 8-month retrospective review of all cataract surgery post-operative Cystoid Macular Oedema (CMO) diagnoses, which is one of the most common complications following surgery[1].

CMO is the retinal thickening of the macula due to a disturbance of the ordinary blood-retinal barrier. This causes an escape of liquid through the perifoveal retinal capillaries which results in an accumulation of fluid within the intracellular spaces of the retina[2].

Purpose

The purpose of the review was to discern whether the change in post-operative drop regime impacted the overall CMO incidence at the clinic.

Methodology

Post-operative '*Drop Regime A*' consisted of **Maxidex 0.1% w/v, eye drops only**.

Post-operative '*Drop Regime B*' consisted of a **combination of both Maxidex 0.1% w/v, eye drops and Acular 0.5% w/v eye drops**.

Data collation involved reviewing every surgical patient throughout the defined period, determining which drops were prescribed and then ascertaining whether these patients had any post-operative complications such as CMO.

- The '*Drop Regime A*' cohort included a total of **1,539 patients**.
- The '*Drop Regime B*' cohort included a total of **1,569 patients**.

Conclusions

The review highlighted that the practice of post-operative use of both Maxidex 0.1% w/v, eye drops and Acular 0.5% w/v eye drops drastically **reduced CMO incidence** following cataract surgery in those patients reviewed. This in turn reduced morbidity for patients.

Acknowledgments

- Review conducted at request of Kaveri Mandal, Consultant Ophthalmologist.
- Patient data has been used in accordance with Newmedica's Privacy Policy which can be found at https://www.newmedica.co.uk/privacy-notice
- I can confirm that the data within has been anonymised however such alterations have not distorted scientific meaning.

Results

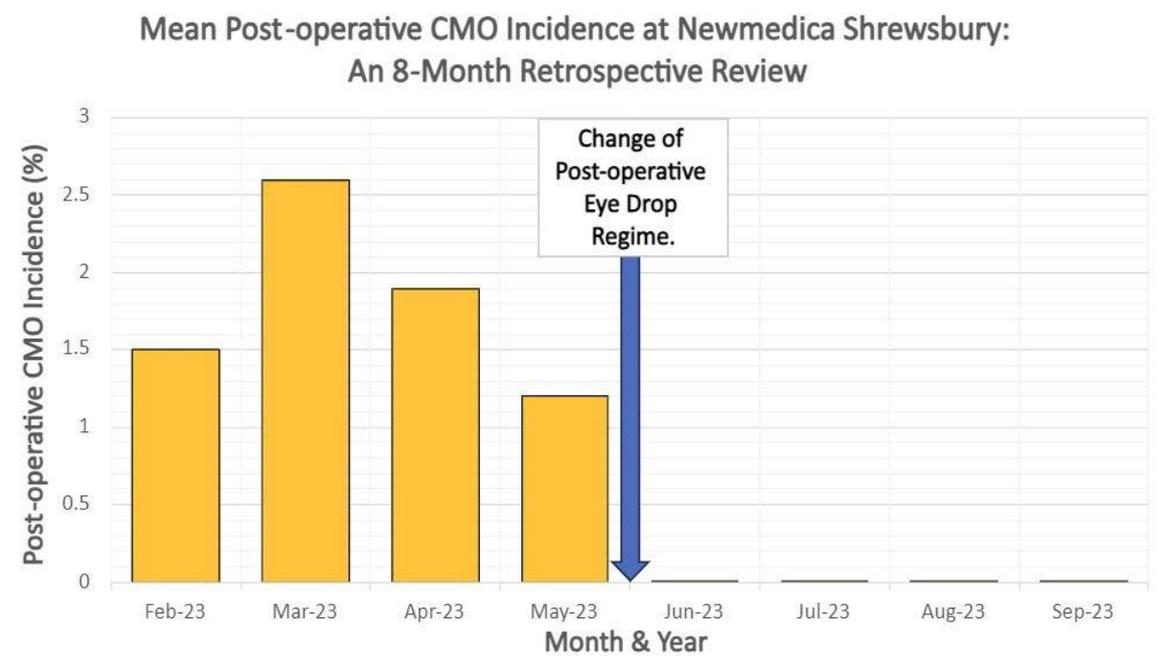

Figure 1 – *Graph presenting the mean post-operative CMO incidence at Newmedica Shrewsbury across each month reviewed.*

After 4-months of data collation for post-operative '*Drop Regime A*' (February 2023 - May 2023), the review concluded that on average, **1 in every 56 patients** were diagnosed with CMO following cataract surgery. Conversely, data collation for post-operative '*Drop Regime B*' (June 2023 - September 2023), concluded that on average, **no patients** were diagnosed with CMO following cataract surgery.

Limitations

- Minimal period of review – Data has only been collated over 8-months.
- Data capture – Assumes data has been captured correctly and in full.
- Community referrals – Assumes that all CMO cases observed during community post-operative review are referred to Newmedica Shrewsbury for treatment.
- Patient compliance – Assumes that all patients adhered to regime as prescribed.

References

1. Miller, K., Choi, J., Patel, A. S., Delmonte, D. W., Baartman, B., Randolph, J. (2023) 'Cataract Surgery Complications', *American Academy of Ophthalmology*. Available at: https://eyewiki.aao.org/Cataract_Surgery_Complications (Accessed: 14 November 2023).

2. Feldman, B. H., Garg, P. G., Shah, V. A., Kim, L. A., Lee, S.Y., Tripathy, K., Hossain, K., Do, D. V., Lim J. I. (2023) 'Cystoid Macular Edema', *American Academy of Ophthalmology*. Available at: https://eyewiki.aao.org/Cystoid_Macular_Edema (Accessed: 14 November 2023).